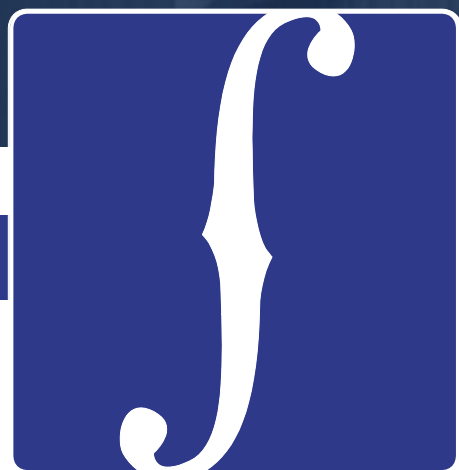




CELLO

Agency Prepayment V2.10

July 2016

1

The model is too fast after 2014

- Model tracking for the post-2014 period is much faster than actual
- The problem is potentially caused by the fact that 2-year HPA is not a good proxy for Mortgage Credit Availability. In particular, other indicators of MCA suggest that this metric was relatively unchanged post-2014 as opposed to the significant pickup in 2-year HPA over this period.
- We need to locate a better proxy for tracking changes in MCA.

2

The model shows too little sensitivity with respect to loan size

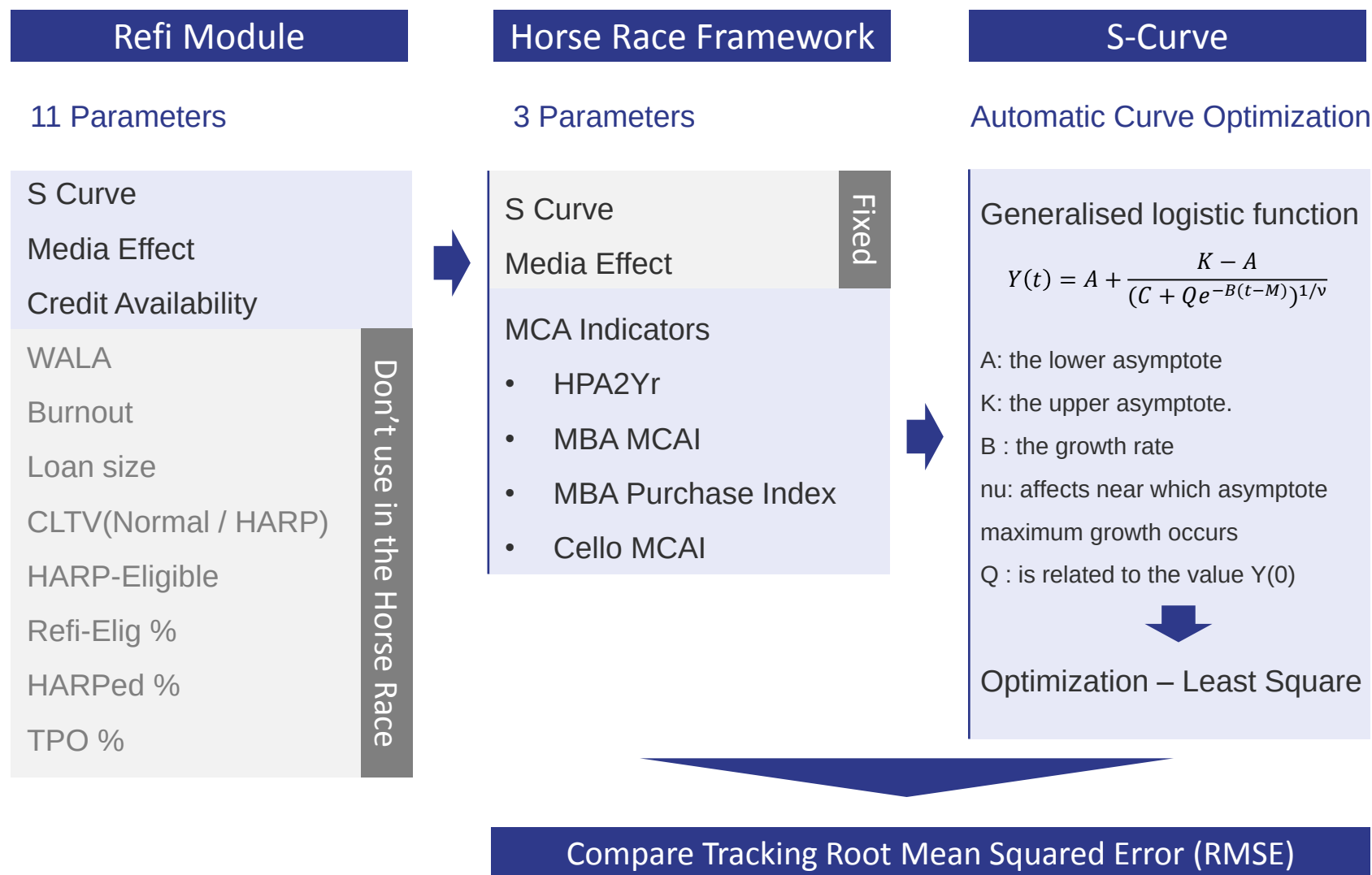
- Refinancing sensitivity is too low for loan sizes in the 50K-300K range
- Improved multiplier for loan size. A future iteration will also take “elbow” effects into consideration

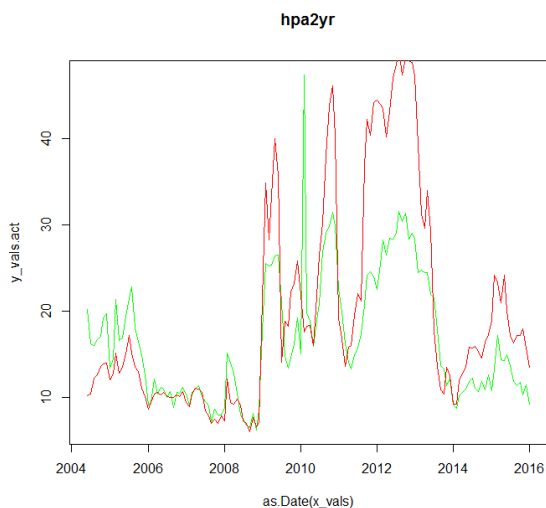
3

The model is too slow for high LTV loans

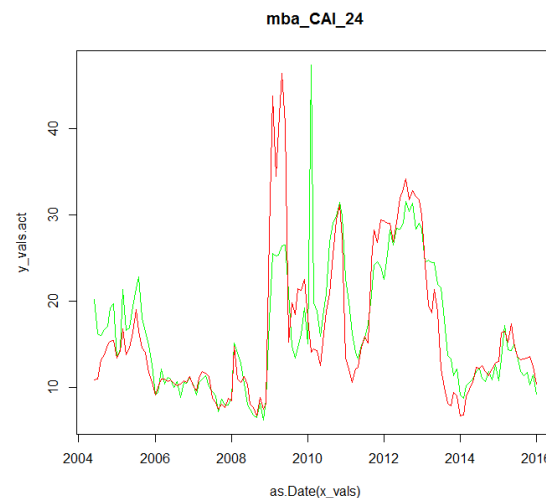
- Tracking for CR pools is too slow
- One potential suspect is turnover projections for CRs (as most of the CR pools are out-of-the-money, their speeds are largely determined by turnover model)
- Revisit turnover model parameters for High LTV loans

To select an alternative proxy for MCA, we compared four other candidates

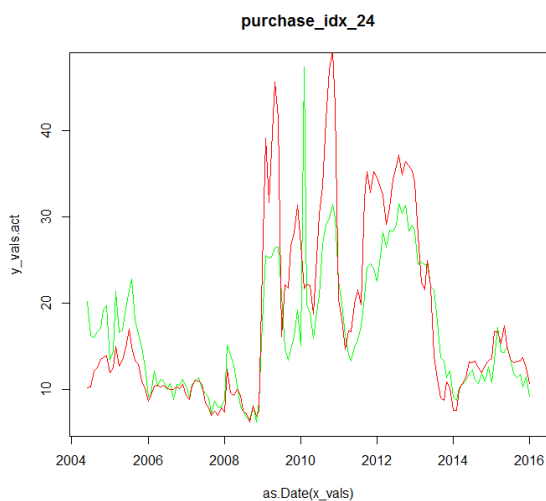




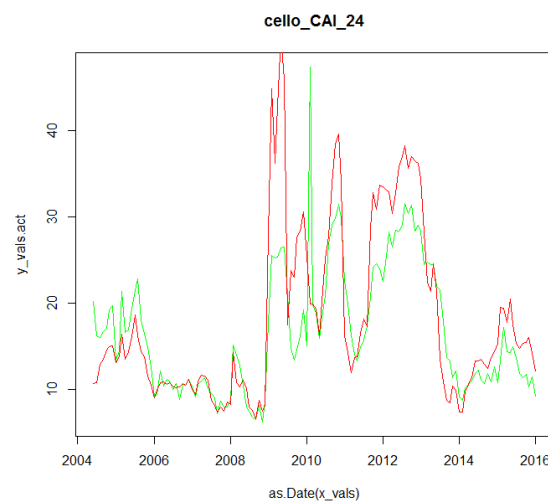
RMSE = $7.9710e-2$



RMSE = $7.7610e-2$



RMSE = $7.8410e-2$



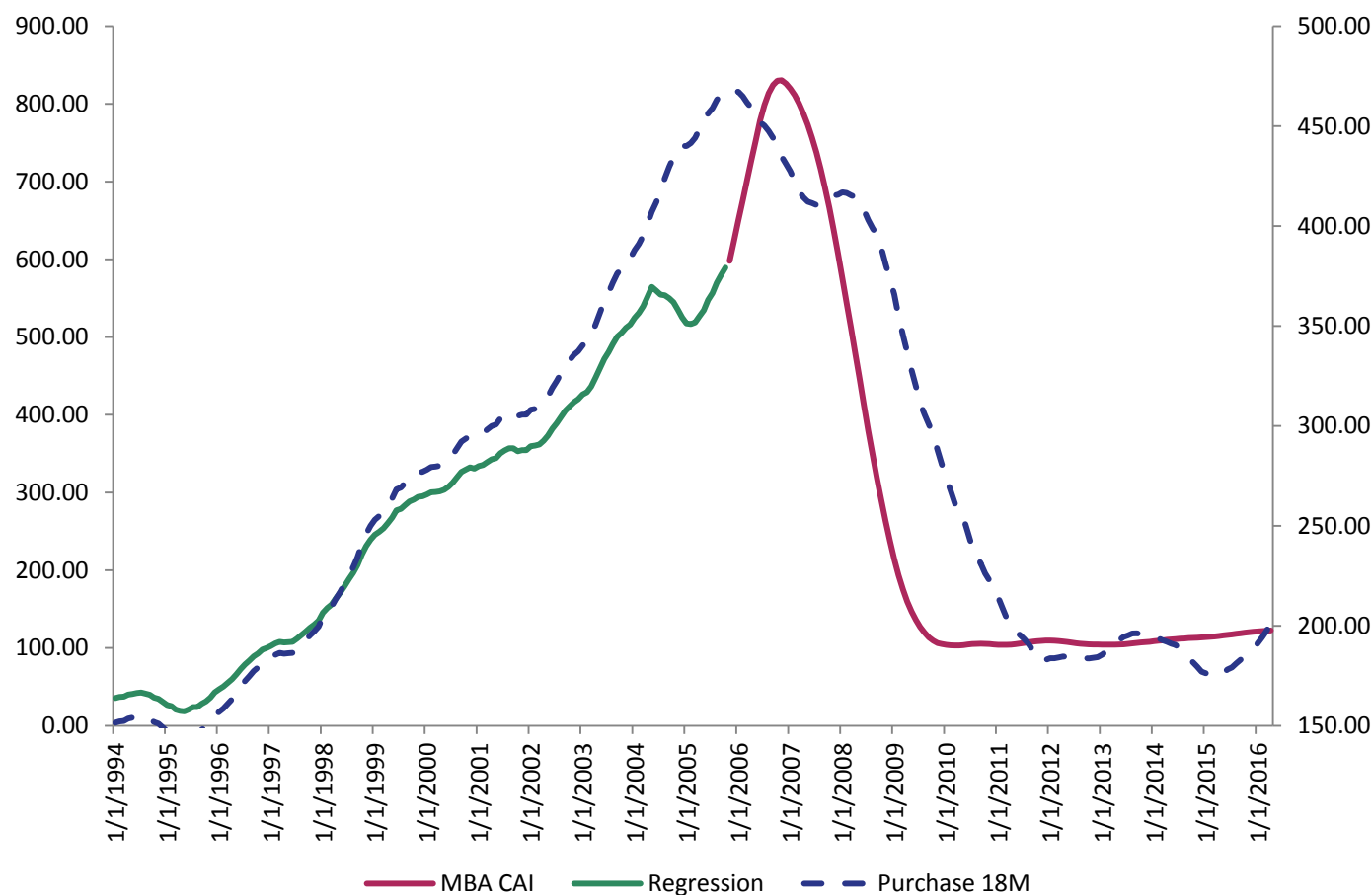
RMSE = $7.8410e-2$

Using the four candidate MCA indicators, we tested the tracking from 2004 to 2016. The total RMSE shows the three new curves have better fits compared with the HPA-2yr index. The MBA CAI has the best performance among the four.

	mba_CAI	purchase_idx	cello_CAI
24 Month	0.0775741	0.0784130	0.0783926
18 Month	0.0770801	0.0787880	0.0777685
12 Month	0.0791511	0.0791555	0.0862411
6 Month	0.0783332	0.0807010	0.1008529

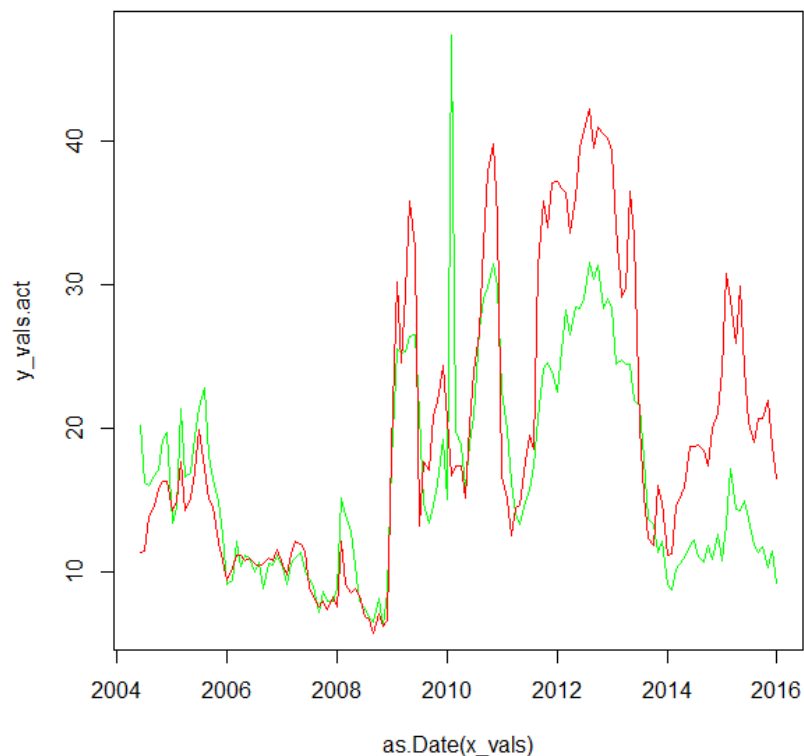
From the RMSE numbers showed above, the best combination is MBA MCAI with an 18 month moving average. All three potential replacements demonstrate reasonable tracking performance when either an 18- or 24-month moving average is used. Performance degrades when shorter (6- or 12-month) time periods are chosen.

As the MBA CAI data is only available after 2004, we need to backfill the data set. Since the next best candidate for the MCA Index is the Purchase Index and there is a significant amount of historical data available on the Purchase Index, we will use trends in the Purchase Application data to backfill the MCAI.

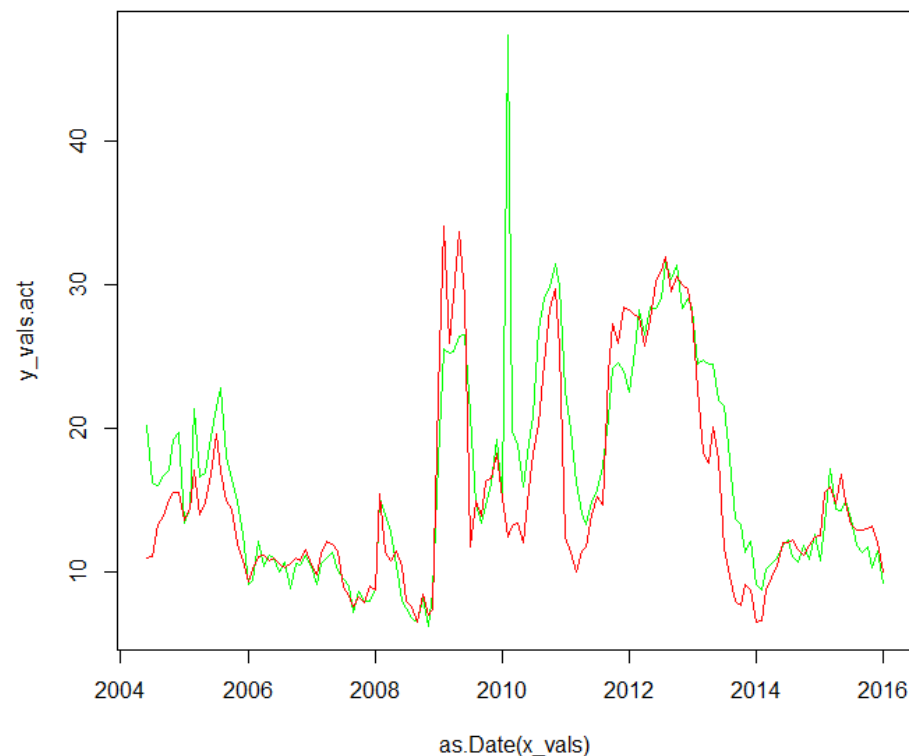


We regress MBA Purchase Applications data against the MBA MCAI from 2005 to 2016 and use the model to backfill history going back to 1994.

Tracking Result Using Current HPA2Yr Model

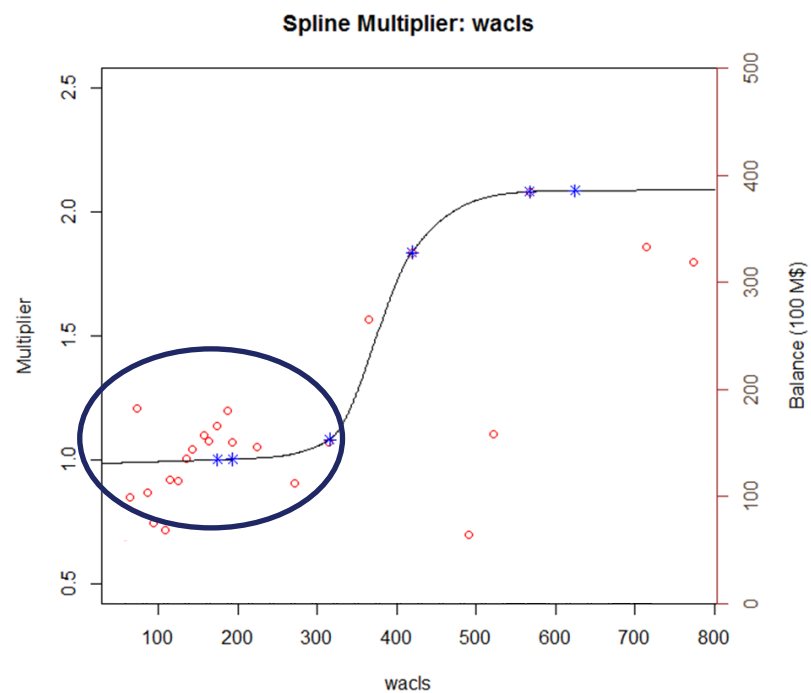


Tracking Result Using MBA MCAI 18 Month

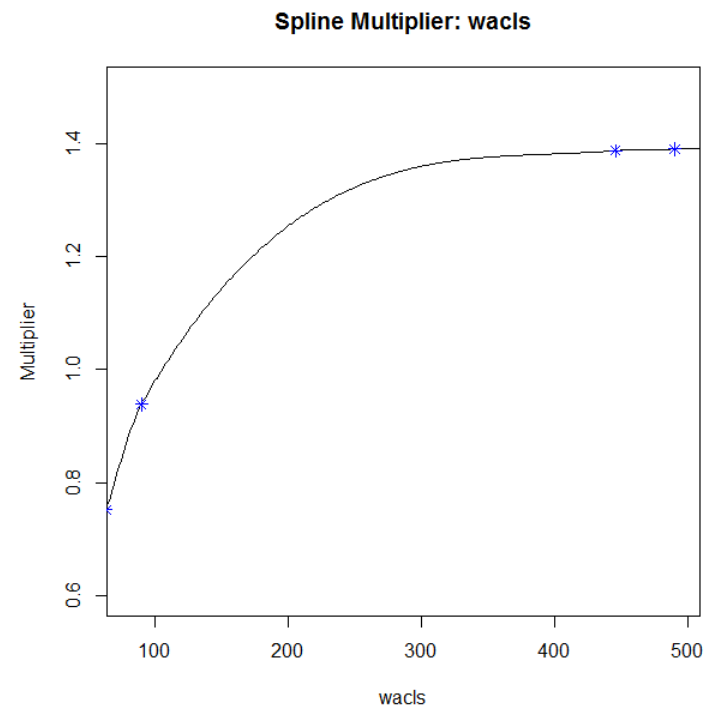


From the charts, we can clearly see a significant improvement in the model fits. In particular, the issue that the model is tracking too fast after 2014 should be addressed.

Loan Size Curve in V2.00



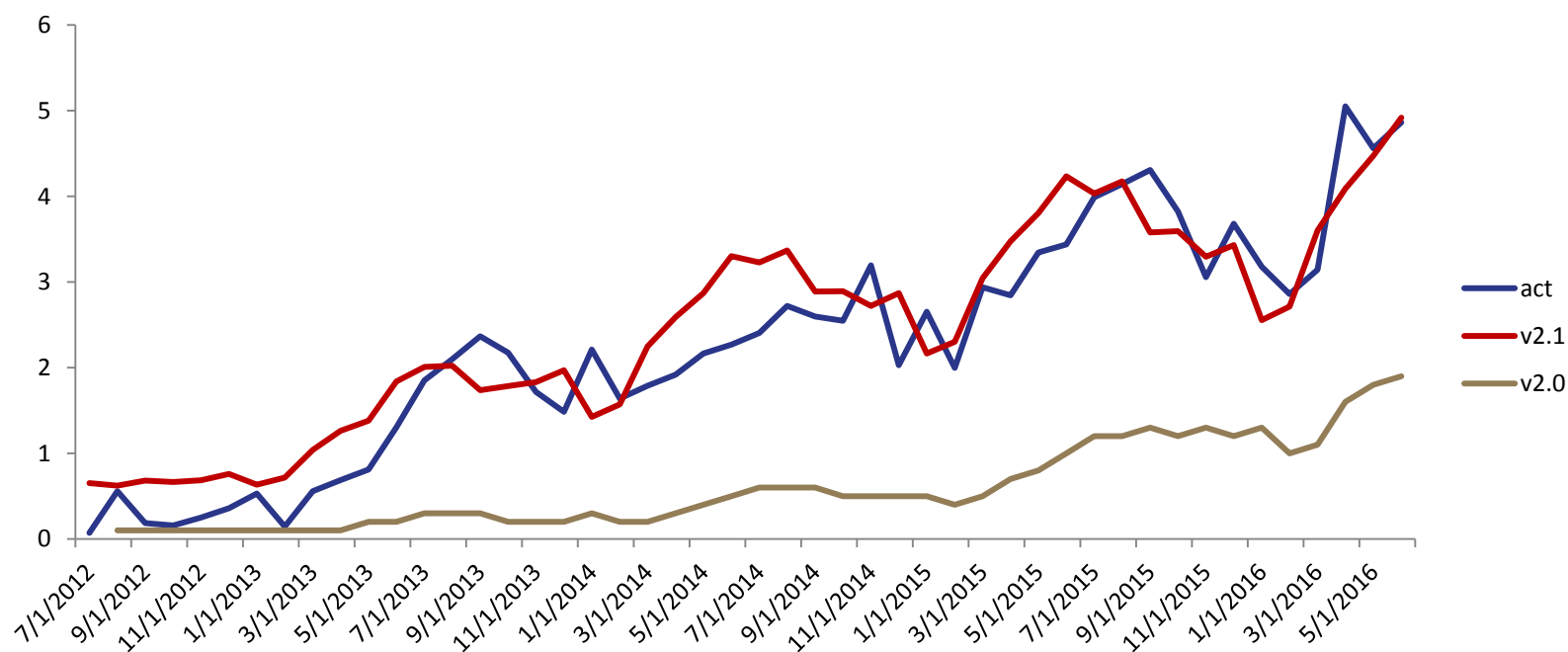
Loan Size Curve in V2.10



In v2.1, we decided to focus on the range between 0-300K, where most of the data falls in. Patching up the curve here should address a significant portion of the sensitivity issues.

In v2.0, the model tracking for high CLTV pools is too slow..

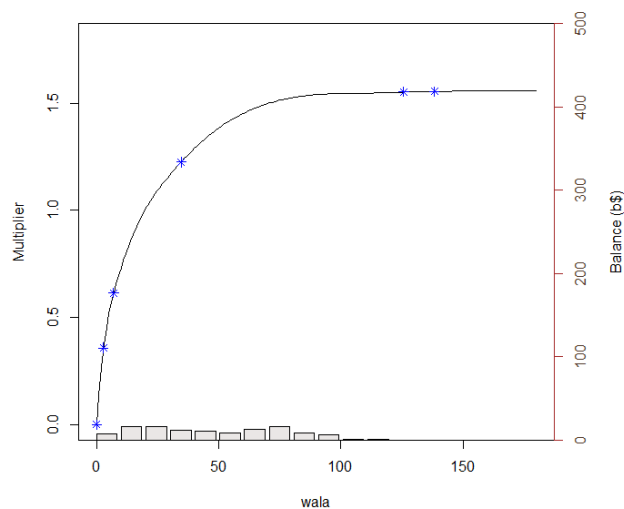
Tracking for Turnover Model (FNU9)



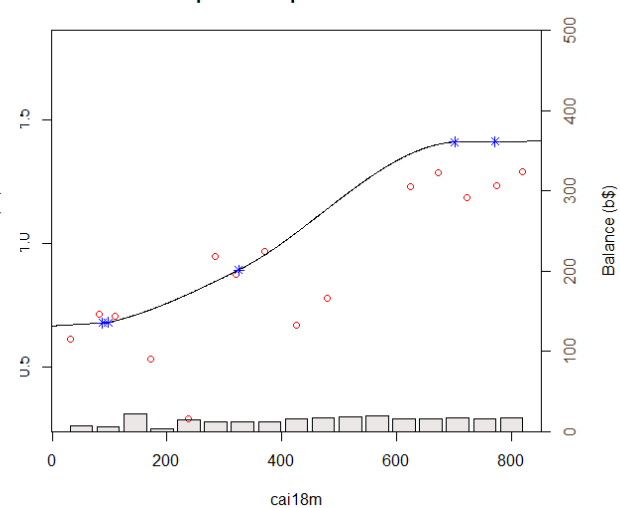
The improvement focused on the higher end of CLTV curve.

As the credit availability parameter is the second in the sequential fitting process. After fitting it, we need to rebuild all other curves after it. WALA is not affected by the update.

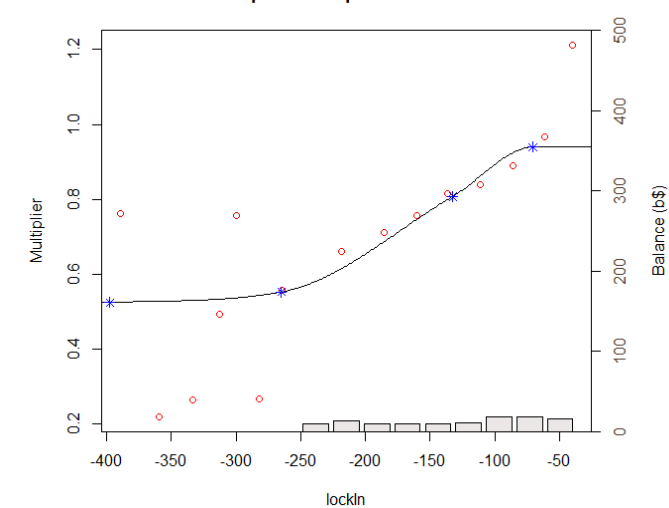
Spline Multiplier: wala



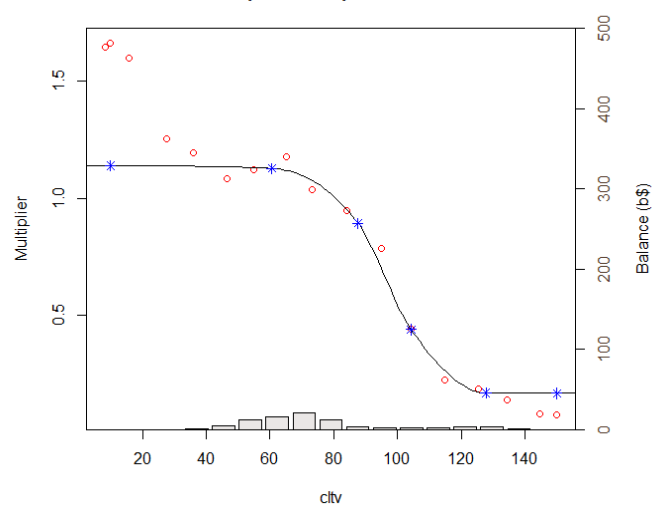
Spline Multiplier: cai18m



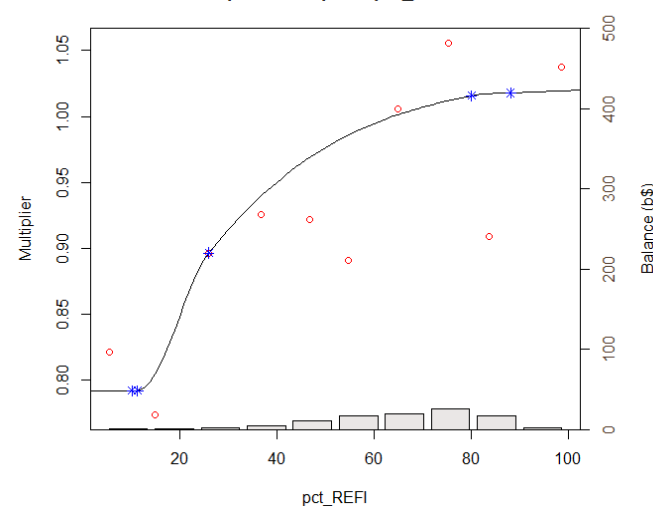
Spline Multiplier: lockIn



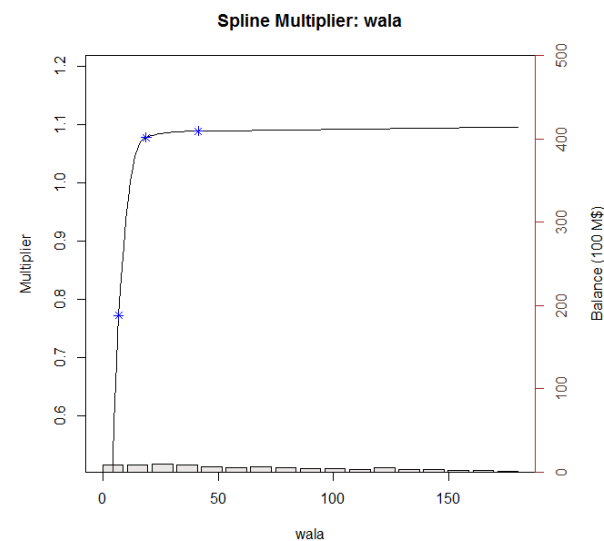
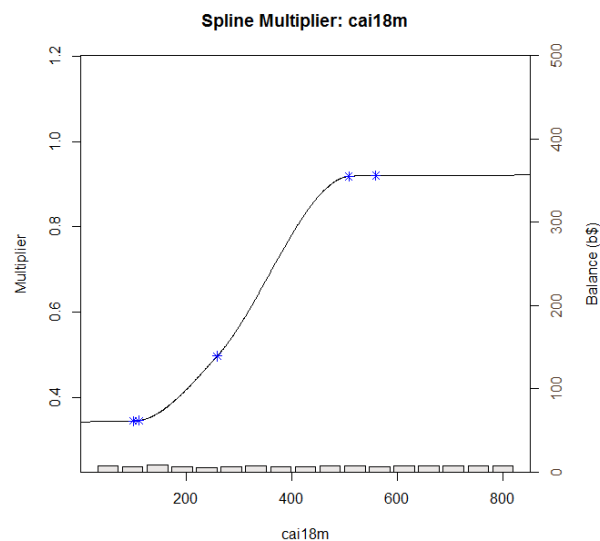
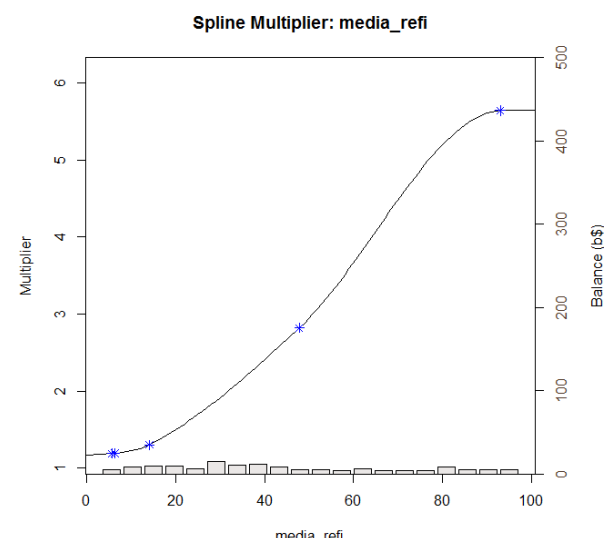
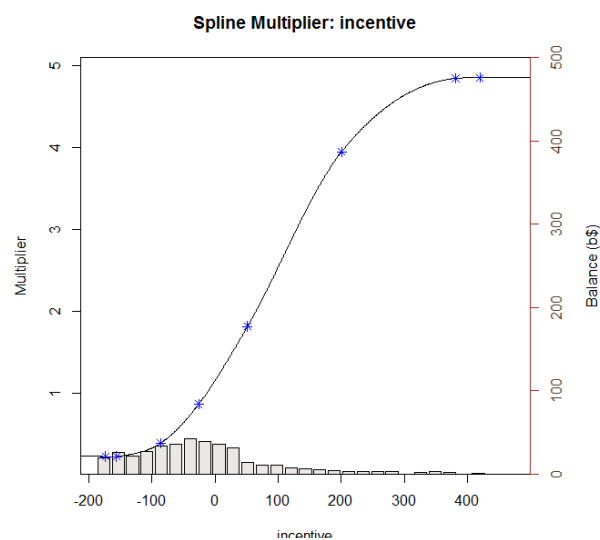
Spline Multiplier: cltv



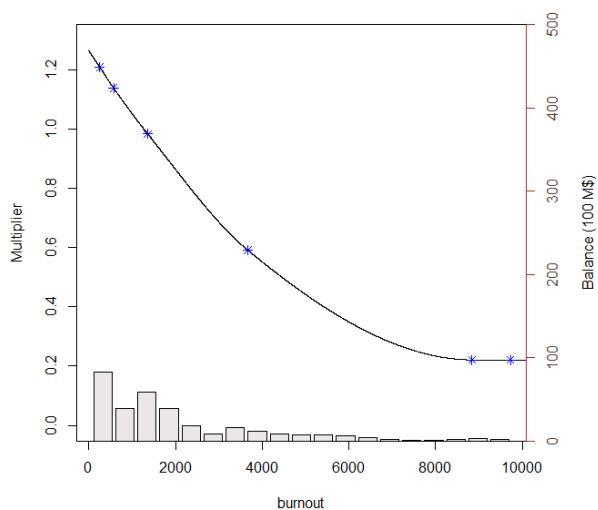
Spline Multiplier: pct_REFI



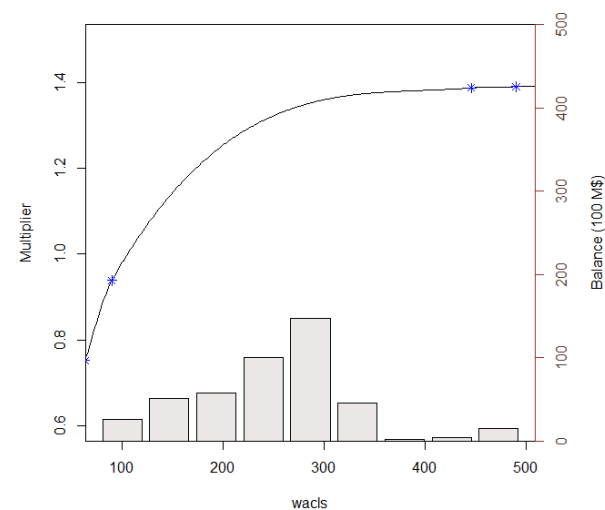
Similar to the Turnover submodel, the credit availability parameter is the third in the sequential fitting process. After fitting it, we need to rebuild all other curves after it. And due to the Turnover model update, the baseline S-curve also needs to be updated.



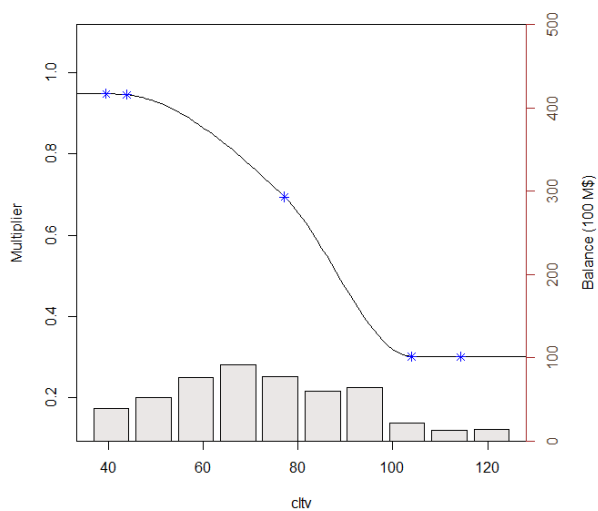
Spline Multiplier: burnout



Spline Multiplier: wacIs



Spline Multiplier: cItv



Spline Multiplier: refiEligPct

